

Solution to Homework 6

① a) $H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5$

H_a : at least one of the population means ($\mu_i, i=1, \dots, 5$) is different from the others.

Since the p-value ($= 6.102 \cdot 10^{-12}$) is smaller than α , we reject the null hypothesis.

c) At least one of the materials is different than the others in terms of how they wear based on their friction test values.

d) $\ell_1 = \frac{\mu_1 + \mu_2 + \mu_3}{3} - \frac{\mu_4 + \mu_5}{2} \rightarrow 2\mu_1 + 2\mu_2 + 2\mu_3 - 3\mu_4 - 3\mu_5$

$\ell_2 = \mu_1 - \frac{\mu_2 + \mu_3}{2} \rightarrow 2\mu_1 - 1\mu_2 - 1\mu_3 + 0\mu_4 + 0\mu_5$

$\ell_3 = \mu_2 - \mu_3 \rightarrow 0\mu_1 + 1\mu_2 - 1\mu_3 + 0\mu_4 + 0\mu_5$

$\ell_4 = \mu_4 - \mu_5 \rightarrow 0\mu_1 + 0\mu_2 + 0\mu_3 + 1\mu_4 - 1\mu_5$

(1-2) $4 - 2 - 2 + 0 + 0 = 0$

(1-3) $0 + 2 - 2 + 0 + 0 = 0$

(1-4) $0 + 0 + 0 - 3 + 3 = 0$

(2-3) $0 - 1 + 1 + 0 + 0 = 0$

(2-4) $0 + 0 + 0 + 0 + 0 = 0$

(3-4) $0 + 0 + 0 + 0 + 0 = 0$

① e)

l_1 : Compare the mean wear for brands produced by Company A with the mean wear for brands produced by Company B.

l_2 : Compare the mean wear for ACME (which is black) with the average mean wear for CHAMP and AJAX. Note that all of these brands are produced by company A.

l_3 : Compare the mean wear for CHAMP and AJAX (both are produced by Company A, and both of them are blue).

l_4 : Compare the mean wear for brands TUFFY and XTRA.

$$f) \text{ SSC}_1 = \frac{(\hat{l}_1)^2}{\sum_{i=1}^5 \frac{a_i^2}{n_i}} = \frac{n(\hat{l}_1)^2}{\sum_{i=1}^5 a_i^2}$$

For contrast l_1 : $a_1 = 2$ $a_2 = 2$ $a_3 = 2$ $a_4 = -3$ $a_5 = -3$

$$\begin{aligned} \hat{l}_1 &= \sum_{i=1}^5 a_i \bar{y}_i = 2(2.675) + 2(2.375) + 2(1.15) \\ &\quad - 3(3.5) - 3(3.375) \\ &= 5.35 + 4.75 + 2.3 - 10.5 - 10.125 \\ &= -8.225 \end{aligned}$$

① f, cont.

$$\hat{l}_1 = -8.225$$

$$n_1 = n_2 = n_3 = n_4 = n_5 = n = 4$$

$$\sum_{i=1}^5 a_i^2 = 2^2 + 2^2 + 2^2 + (-3)^2 + (-3)^2 = 4 + 4 + 4 + 9 + 9 = 30$$

$$SSC_1 = \frac{4(-8.225)^2}{30} = \underline{\underline{9.02}}$$

$$SSC_2 = \frac{n(\hat{l}_2)^2}{\sum_{i=1}^n a_i^2}$$

For contrast l_2 : $a_1=2$ $a_2=-1$ $a_3=-1$ $a_4=0$ $a_5=0$

$$\begin{aligned}\hat{l}_2 &= \sum_{i=1}^5 a_i \bar{Y}_i = 2(2.675) - 1(2.375) - 1(1.15) \\ &= 5.35 - 2.375 - 1.15 = 1.825\end{aligned}$$

$$\sum_{i=1}^n a_i^2 = 2^2 + (-1)^2 + (-1)^2 = 4 + 1 + 1 = 6$$

$$SSC_2 = \frac{4(1.825)^2}{6} = \underline{\underline{2.22}}$$

$$SSC_3 = \frac{n(\hat{l}_3)^2}{\sum_{i=1}^n a_i^2}$$

For contrast l_3 : $a_1=0$ $a_2=1$ $a_3=-1$ $a_4=0$ $a_5=0$

$$\hat{l}_3 = \sum_{i=1}^5 a_i \bar{Y}_i = 1(2.375) - 1(1.15) = 1.225$$

$$\sum_{i=1}^5 a_i^2 = 1^2 + (-1)^2 = 2$$

$$SSC_3 = \frac{4(1.225)^2}{2} = \underline{\underline{3.00}}$$

① f_1 cont.

$$SSC_4 = \frac{n(\hat{l}_4)^2}{\sum_{i=1}^5 a_i^2}$$

For contrast l_4 : $a_1=0$ $a_2=0$ $a_3=0$ $a_4=1$ $a_5=-1$

$$\hat{l}_4 = \sum_{i=1}^5 a_i \bar{y}_i = 1(3.5) - 1(3.375) = 0.125$$

$$\sum_{i=1}^5 a_i^2 = 1^2 + (-1)^2 = 2$$

$$SSC_4 = \frac{4(0.125)^2}{2} = 0.03125$$

$$SSC_1 + SSC_2 + SSC_3 + SSC_4 =$$

$$9.02 + 2.22 + 3.00 + 0.03 = 14.27$$

$$= SS_{\text{Treatment}} = SS_{\text{material}} \quad \checkmark$$

$$g) \quad t = \frac{\hat{l}}{\sqrt{V(\hat{l})}}$$

$$V(\hat{l}) = s_w^2 \cdot \sum_{i=1}^5 \frac{a_i^2}{n_i}$$

$$n_1 = n_2 = n_3 = n_4 = n_5 = n$$

$$s_w^2 = MSE = 0.0235$$

① g) Cont.

Contrast 1:
 $\hat{\ell}_1 = -8.225$

$$V(\hat{\ell}_1) = 0.0235 \cdot \left(\frac{30}{4}\right) = 0.17625$$

$$t = \frac{-8.225}{\sqrt{0.17625}} = -19.59$$

$$t_{0.025, 15} = 2.131$$

Since $|-19.59| = 19.59 > 2.131$, we reject H_0 at $\alpha=0.05$ level.

We conclude that there is a difference between the mean wear for brands produced by Company A and the mean wear for brands produced by Company B.

Contrast 2:

$$\hat{\ell}_2 = 1.825$$

$$V(\hat{\ell}_2) = 0.0235 \left(\frac{6}{4}\right) = 0.03525$$

$$t = \frac{1.825}{\sqrt{0.03525}} = 9.72$$

$$t_{0.025, 15} = 2.131$$

Since $9.72 > 2.131$ we reject H_0 , and conclude that there is a difference in mean wear for brands that are blue and black in Company A.

① g) cont.

Contrast 3:

$$\hat{l}_3 = 1.225$$

$$V(\hat{l}_3) = 0.0235 \left(\frac{2}{4} \right) = 0.01175$$

$$t = \frac{1.225}{\sqrt{0.01175}} = 11.30$$

$$t_{0.025, 115} = 2.131$$

Since $11.30 > 2.131$ we reject H_0 , and conclude that there is a difference between brands CHAMP and AJAX in terms of wear.

Contrast 4:

$$\hat{l}_4 = 0.125$$

$$V(\hat{l}_4) = 0.0235 \left(\frac{2}{4} \right) = 0.01175$$

$$t = \frac{0.125}{\sqrt{0.01175}} = 1.1532$$

$$t_{0.025, 115} = 2.131$$

Since $1.1532 < 2.131$ we fail to reject H_0 , and conclude that there is no difference between TUFFY and XTRA in terms of wear.

① h)

$$F = \frac{SSC}{MSE}$$

$$MSE = 0.0235$$

$$F_{0.05, 1, 15} = 4.54$$

For contrast 1:

$$F = \frac{SSC_1}{MSE} = \frac{9.02}{0.0235} = 383.83$$

Since $383.83 > 4.54$ we reject H_0 , and conclude same as in ① g) for contrast 1.

For contrast 2:

$$F = \frac{SSC_2}{MSE} = \frac{2.22}{0.0235} = 94.47$$

Since $94.47 > 4.54$ we reject H_0 , and conclude same as in ① g) for contrast 2.

For contrast 3:

$$F = \frac{SSC_3}{MSE} = \frac{3.00}{0.0235} = 127.66$$

Since $127.66 > 4.54$ we reject H_0 , and conclude same as in ① g) for contrast 3.

For contrast 4:

$$F = \frac{SSC_4}{MSE} = \frac{0.03125}{0.0235} = 1.3298$$

Since $1.3298 < 4.54$ we fail to reject H_0 , and conclude same as in ① g) for contrast 4.

$$\textcircled{2} i, \mu_1 + 3\mu_2 - 4\mu_3$$

The linear combination is a linear contrast if $\sum_i a_i = 0$

$$a_1 = 1 \quad a_2 = 3 \quad a_3 = -4$$

$$\sum a_i = 1 + 3 - 4 = 0 \quad \checkmark \quad \text{It is a linear contrast.}$$

$$ii, 0.3\mu_1 + 0.5\mu_2 + 0.1\mu_3 + 0.1\mu_4$$

$$a_1 = 0.3 \quad a_2 = 0.5 \quad a_3 = 0.1 \quad a_4 = 0.1$$

$$\sum a_i = 0.3 + 0.5 + 0.1 + 0.1 = 1 \neq 0$$

It is not a linear contrast.

$$iii, \frac{1}{3}(\mu_1 + \mu_2 + \mu_3) - \mu_4$$

$$a_1 = 1 \quad a_2 = 1 \quad a_3 = 1 \quad a_4 = -3$$

$$\sum a_i = 1 + 1 + 1 - 3 = 0 \quad \checkmark$$

$$\text{Or } a_1 = \frac{1}{3} \quad a_2 = \frac{1}{3} \quad a_3 = \frac{1}{3} \quad a_4 = -1$$

$$\sum a_i = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} - 1 = 0 \quad \checkmark$$

It is a linear contrast.

③ $\alpha_E = 0.05$

Then $\alpha_I = \frac{0.05}{4} = 0.0125$

Then the critical value is $F_{0.0125, 1, 15} = 8.68$
(I used $F_{0.01, 1, 15}$)

$F_1 = 383.83 \rightarrow \text{reject } H_0$

$F_2 = 94.47 \rightarrow \text{reject } H_0$

$F_3 = 127.66 \rightarrow \text{reject } H_0$

$F_4 = 1.3298 \rightarrow \text{fail to reject } H_0$

} Same conclusions
as in ① h.

④ a)

Fisher's protected LSD:

$\bar{Y}_{1.} = 2.675$

$\bar{Y}_{2.} = 2.375$

$\bar{Y}_{3.} = 1.15$

$\bar{Y}_{4.} = 3.5$

$\bar{Y}_{5.} = 3.375$

$\rightarrow$ rank the
sample
means
from
lowest
to highest

Brand	3	2	1	5	4
$Y_{i.}$	1.15	2.375	2.675	3.375	3.5
New ranking	1	2	3	4	5

$\alpha = 0.05$

$LSD = t_{\alpha/2, 15} \sqrt{\frac{2MSE}{n}}$

$= 2.131 \sqrt{\frac{2 \cdot 0.0235}{4}}$

$= \underline{\underline{0.23}}$

Using the new ranking:

$\bar{Y}_{5.} - \bar{Y}_{1.} = 2.35 > LSD$

$\bar{Y}_{4.} - \bar{Y}_{1.} = 2.225 > LSD$

$\bar{Y}_{3.} - \bar{Y}_{1.} = 1.525 > LSD$

$\bar{Y}_{2.} - \bar{Y}_{1.} = 1.225 > LSD$

$$\bar{y}_{5.} - \bar{y}_{2.} = 1.125 > \text{LSD}$$

$$\bar{y}_{4.} - \bar{y}_{2.} = 1 > \text{LSD}$$

$$\bar{y}_{3.} - \bar{y}_{2.} = 0.3 > \text{LSD}$$

$$\bar{y}_{5.} - \bar{y}_{3.} = 0.825 > \text{LSD}$$

$$\bar{y}_{4.} - \bar{y}_{3.} = 0.7 > \text{LSD}$$

$$\bar{y}_{5.} - \bar{y}_{4.} = 0.125 < \text{LSD}$$

Brand with new ranking	1	2	3	4	5
Comparison					

Other than Tukey and XTRA the brands are significantly different from each other.

b) Tukey's W:

$$t = 5$$

$$v = 15$$

$$\alpha_E = 0.05$$

$$n = 4$$

$$q_{0.05}(5, 15) = 4.37$$

$$W = q_{0.05}(5, 15) \cdot \sqrt{\frac{S_w^2}{n}} = 4.37 \cdot \sqrt{\frac{0.0235}{4}}$$

$$= \underline{0.335}$$

We just substitute W for LSD, and use the comparisons from part a,

According to Tukey's W we have the following

1 2 3 4 5

μ_2 and μ_3 are significantly different from μ_4 and μ_5 .
 μ_1 is different from all of $\mu_2, \mu_3, \mu_4, \mu_5$.

④ c, Scheffe's S:

	$\hat{\ell}$	$V(\hat{\ell})$	S	Conclusion
Contrast 1	-8.225	0.17625	1.47	Significant diff.
Contrast 2	1.825	0.03525	0.66	Significant diff.
Contrast 3	1.225	0.01175	0.38	Significant diff.
Contrast 4	0.125	0.01175	0.38	Significant diff.

$$S = \sqrt{\hat{V}(\hat{\ell})} \sqrt{(t-1) F_{\alpha, df_1, df_2}}$$

$$F_{0.05, 4, 15} = 3.06$$

$$S_1 = \sqrt{0.17625} \cdot \sqrt{4 \cdot 3.06} = 1.468775$$

$$S_2 = \sqrt{0.03525} \cdot \sqrt{4 \cdot 3.06} = 0.6569$$

$$S_3 = \sqrt{0.01175} \cdot \sqrt{4 \cdot 3.06} = 0.3792$$

$$S_4 = \sqrt{0.01175} \cdot \sqrt{4 \cdot 3.06} = 0.3792$$

For conclusion compare $\hat{\ell}$ and S.

If $|\hat{\ell}| > S$, there is a significant diff.